ElderCareAI Core Database Dictionary

Updated 7 November 2026

This dictionary describes the 18 main tables supporting ElderCareAI patient care, Galaxy Watch monitoring, GPS tracking, care plans, alerts, Elle and safe zones. It documents 245 fields from the deployed Supabase PostgreSQL schema.

The core tables are shown below. Supabase Auth and supporting onboarding, provider-link, delivery-queue, internal state and legacy tables are outside this document. Excluding a table here does not remove it from the database.

PK means primary key; FK means a declared foreign key. U means uniqueness: Yes is individually unique, No has no listed uniqueness rule, and * is part of a composite or conditional unique index described below its table. Nullable Yes means the database permits NULL; it does not mean a measured value of zero.

TIMESTAMPTZ stores a timestamp with a time-zone-aware instant. NUMERIC precision and scale are shown exactly. Field names remain database identifiers. All listed core tables have row-level security enabled; backend-managed writes and ownership rules still apply.

Schema reference migration 20261107000000

Table 23
DATA DICTIONARY OF CAREGIVERS

public.caregivers. Stores caregiver identity, contact information and profile details linked to the authenticated account.

Column Name	Data Type	U	Description	Nullable	PK	FK
age	INTEGER	No	Age in years.	Yes	No	No
created_at	TIMESTAMPTZ	No	Date and time the database record was created.	No	No	No
email	TEXT	No	Caregiver's account email address.	Yes	No	No
full_name	TEXT	No	Caregiver's full name shown across the app.	No	No	No
id	UUID	Yes	Unique identifier for this caregiver.	No	Yes	Yes
phone	TEXT	No	Caregiver's contact telephone number.	Yes	No	No
photo_url	TEXT	No	URL of the caregiver's profile photo.	Yes	No	No
sex	TEXT	No	Caregiver's recorded sex.	Yes	No	No
updated_at	TIMESTAMPTZ	No	Date and time the record was last updated.	No	No	No

Foreign keys: id references auth.users(id).

Access: row-level security policies govern caregiver operations.

Table 24
DATA DICTIONARY OF ELDERLY PROFILES

public.elderly_profiles. Stores the older adult profile, health background and emergency contacts managed by a caregiver.

Column Name	Data Type	U	Description	Nullable	PK	FK
age	INTEGER	No	Older adult's age in years.	Yes	No	No
blood_type	TEXT	No	Recorded blood group and Rh type.	Yes	No	No
caregiver_id	UUID	No	Caregiver account associated with this record.	No	No	Yes
created_at	TIMESTAMPTZ	No	Date and time the database record was created.	No	No	No
date_of_birth	DATE	No	Older adult's date of birth.	Yes	No	No
elderly_id	UUID	Yes	Patient profile associated with this record.	No	Yes	No
emergency_contact	TEXT	No	Name of the emergency contact.	Yes	No	No
emergency_contact_phone	TEXT	No	Emergency contact's telephone number.	Yes	No	No
full_name	TEXT	No	Older adult's full name shown in care views.	No	No	No
gender	TEXT	No	Older adult's recorded gender.	Yes	No	No
height_cm	NUMERIC(5,2)	No	Recorded height in centimeters.	Yes	No	No
medical_conditions	TEXT	No	Recorded medical conditions or history.	Yes	No	No
medications	TEXT	No	Medication information entered in the profile.	Yes	No	No
photo_url	TEXT	No	URL of the older adult's profile photo.	Yes	No	No

Column Name	Data Type	U	Description	Nullable	PK	FK
updated_at	TIMESTAMPTZ	No	Date and time the record was last updated.	No	No	No
weight_kg	NUMERIC(5,2)	No	Recorded weight in kilograms.	Yes	No	No

Foreign keys: caregiver_id references public.caregivers(id).

Access: row-level security policies govern caregiver operations.

Table 25
DATA DICTIONARY OF WEARABLE DEVICES

public.wearable_devices. Registers the Galaxy Watch paired to a patient and protects its device access credential. Device registration is managed by the backend.

Column Name	Data Type	U	Description	Nullable	PK	FK
caregiver_id	UUID	No	Caregiver account associated with this record.	No	No	Yes
device_name	TEXT	No	Display name reported for the paired watch.	No	No	No
elderly_id	UUID	*	Patient profile associated with this record.	No	No	Yes
expires_at	TIMESTAMPTZ	No	Time after which the watch credential is no longer accepted.	No	No	No
id	UUID	Yes	Unique identifier assigned to the paired Galaxy Watch.	No	Yes	No
last_history_at	TIMESTAMPTZ	No	Time of the latest accepted Health history import for this watch.	Yes	No	No
last_seen_at	TIMESTAMPTZ	No	Latest time an authenticated request was received from the watch.	Yes	No	No
paired_at	TIMESTAMPTZ	No	Time this watch was paired with the patient account.	No	No	No
revoked_at	TIMESTAMPTZ	No	Time the watch registration was revoked, if disconnected.	Yes	No	No
token_hash	TEXT	Yes	Unique hash of the watch access credential; not the raw credential.	No	No	No

Foreign keys: caregiver_id references auth.users(id); elderly_id references public.elderly_profiles(elderly_id).

Unique combination: elderly_id only when (revoked_at IS NULL).

Unique field: token_hash.

The unique index on elderly_id applies only while revoked_at is NULL. Historical revoked registrations can remain for the same patient.

The raw watch access token is not stored in this table. Backend functions validate the stored hash and credential expiry.

Access: backend managed; direct authenticated table access is not granted.

Table 26
DATA DICTIONARY OF WEARABLE LIVE STATE

public.wearable_live_state. Stores the latest watch telemetry for the live dashboard. Each vital and GPS position retains its own measurement time; a row is not evidence that all sensors measured simultaneously.

Column Name	Data Type	U	Description	Nullable	PK	FK
accuracy_m	DOUBLE PRECISION	No	Reported horizontal position accuracy in meters.	Yes	No	No
battery_percent	INTEGER	No	Watch battery level from 0 to 100 percent.	Yes	No	No
device_id	UUID	No	Registered watch associated with this record.	No	No	Yes
elderly_id	UUID	Yes	Patient identifier and primary key; one live-state row per patient.	No	Yes	Yes
gps_measured_at	TIMESTAMPTZ	No	Original measurement time of the displayed location.	Yes	No	No
gps_status	TEXT	No	Human-readable GPS availability or positioning status.	Yes	No	No
heart_rate_bpm	DOUBLE PRECISION	No	Heart-rate measurement in beats per minute.	Yes	No	No
heart_rate_measured_at	TIMESTAMPTZ	No	Original measurement time of the heart-rate value.	Yes	No	No
hrv_measured_at	TIMESTAMPTZ	No	Original measurement time of the HRV result.	Yes	No	No
hrv_rmssd_ms	DOUBLE PRECISION	No	Heart-rate variability calculated as RMSSD, in milliseconds.	Yes	No	No

Column Name	Data Type	U	Description	Nullable	PK	FK
latitude	DOUBLE PRECISION	No	Latitude in decimal degrees, from -90 to 90.	Yes	No	No
location_source	TEXT	No	Position provider: watch_gps, fused or network.	Yes	No	No
longitude	DOUBLE PRECISION	No	Longitude in decimal degrees, from -180 to 180.	Yes	No	No
monitoring	BOOLEAN	No	Whether the watch reports that monitoring is running.	No	No	No
received_at	TIMESTAMPTZ	No	Time the backend received the watch telemetry.	No	No	No
sensor_status	JSONB	No	JSON object containing availability or collection status for each sensor.	No	No	No
skin_temp_celsius	DOUBLE PRECISION	No	Wrist skin temperature in degrees Celsius; not core body temperature.	Yes	No	No
skin_temp_measured_at	TIMESTAMPTZ	No	Original measurement time of the wrist skin temperature.	Yes	No	No
sleep_hours	DOUBLE PRECISION	No	Recorded sleep duration in hours, imported after Health history sync.	Yes	No	No
sleep_measured_at	TIMESTAMPTZ	No	Measurement time associated with the imported sleep duration.	Yes	No	No
spo2_measured_at	TIMESTAMPTZ	No	Original measurement time of the oxygen reading.	Yes	No	No
spo2_percent	DOUBLE PRECISION	No	Blood-oxygen saturation reading as a percentage.	Yes	No	No
status_at	TIMESTAMPTZ	No	Original timestamp of the watch monitoring and battery status.	No	No	No
steps_count	INTEGER	No	Step total measured during watch monitoring periods.	Yes	No	No
steps_measured_at	TIMESTAMPTZ	No	Original measurement time of the step total.	Yes	No	No
stress_baseline_rmssd	DOUBLE PRECISION	No	Personal HRV reference used to derive the experimental stress index.	Yes	No	No
stress_reference_count	INTEGER	No	Number of reference measurements used for the stress estimate.	No	No	No
stress_score	INTEGER	No	Experimental HRV-based stress index from 0 to 100; not medical grade.	Yes	No	No

Foreign keys: device_id references public.wearable_devices(id); elderly_id references public.elderly_profiles(elderly_id).

This table supports the live display. It does not replace the individual measurement records in vital_sign_logs or the route points in watch_location_points.

Blank values or older per-metric timestamps must not be interpreted as fresh readings from every sensor.

Access: caregiver reads are subject to patient ownership policies; writes use backend functions or triggers.

Table 27
DATA DICTIONARY OF VITAL SIGN LOGS

public.vital_sign_logs. Keeps timestamped vital measurements and imported sleep or oxygen history. Current Galaxy Watch uploads and Health history imports both use source galaxy_watch; measurement_origin distinguishes their collection paths.

Column Name	Data Type	U	Description	Nullable	PK	FK
active_minutes	INTEGER	No	Legacy optional activity-duration field; current watch uploads do not populate it.	Yes	No	No
ai_risk_score	NUMERIC(5,2)	No	Optional stored indicative risk score; not a clinical risk assessment.	Yes	No	No
elderly_id	UUID	*	Patient profile associated with this record.	No	No	Yes
heart_rate_bpm	NUMERIC(6,2)	No	Heart-rate measurement in beats per minute.	Yes	No	No
hrv_rmssd_ms	NUMERIC	No	Heart-rate variability calculated as RMSSD, in milliseconds.	Yes	No	No
id	UUID	Yes	Unique identifier for a saved vital measurement or imported history record.	No	Yes	No

Column Name	Data Type	U	Description	Nullable	PK	FK
is_superseded	BOOLEAN	No	Whether a newer imported revision replaced this record in current history.	No	No	No
measurement_origin	TEXT	No	Collection path: watch_sensor or samsung_health.	Yes	No	No
measurement_times	JSONB	No	JSON object mapping metric names to their original measurement timestamps.	No	No	No
overall_status	TEXT	No	Stored app status field; not a diagnosis or a clinical classification.	Yes	No	No
recorded_at	TIMESTAMPTZ	No	Original measurement time used to order this event in saved history.	No	No	No
skin_temp_celsius	NUMERIC(5,2)	No	Wrist skin temperature in degrees Celsius; not core body temperature.	Yes	No	No
sleep_history	JSONB	No	JSON array of sleep-session details; empty for records without sleep-session data.	No	No	No
sleep_hours	NUMERIC(5,2)	No	Recorded sleep duration in hours, imported after Health history sync.	Yes	No	No
source	TEXT	No	Data-source family. Current watch and Health history paths explicitly write galaxy_watch.	Yes	No	No
source_device_id	TEXT	*	Provider device identifier selected for the Health history import.	Yes	No	No
source_group_key	TEXT	No	Provider group or session key linking related imported measurements.	Yes	No	No
source_record_key	TEXT	*	Provider record key used to identify an imported measurement.	Yes	No	No
source_revision_at	TIMESTAMPTZ	*	Provider modification time identifying the imported record revision.	Yes	No	No
spo2_percent	NUMERIC(5,2)	No	Blood-oxygen saturation reading as a percentage.	Yes	No	No
steps_count	INTEGER	No	Step total measured during watch monitoring periods.	Yes	No	No
stress_baseline_rmssd	DOUBLE PRECISION	No	Personal HRV reference used to derive the experimental stress index.	Yes	No	No
stress_model_version	TEXT	No	Version identifier of the experimental stress-scoring method.	Yes	No	No
stress_reference_count	INTEGER	No	Number of reference measurements used for the stress estimate.	Yes	No	No
stress_score	INTEGER	No	Experimental HRV-based stress index from 0 to 100; not medical grade.	Yes	No	No
synced_at	TIMESTAMPTZ	No	Time this measurement was saved or synchronized to the backend.	Yes	No	No
watch_device_id	UUID	*	Registered watch that produced or supplied this measurement.	Yes	No	Yes
watch_measured_at	TIMESTAMPTZ	*	Original event timestamp used in watch measurement deduplication.	Yes	No	No
watch_metric	TEXT	*	Metric represented by the saved watch or imported measurement event.	Yes	No	No
watch_sample_id	UUID	*	Identifier of the delivered watch sensor event, used to deduplicate retries.	Yes	No	No

Foreign keys: elderly_id references public.elderly_profiles(elderly_id); watch_device_id references public.wearable_devices(id).

Unique combination: elderly_id, source_device_id, source_record_key, source_revision_at only when (measurement_origin = 'samsung_health'::text).

Unique combination: watch_device_id, watch_metric, watch_measured_at only when (measurement_origin IS DISTINCT FROM 'samsung_health'::text).

Unique combination: watch_device_id, watch_sample_id.

A sparse row can contain one measured vital and NULL in other vital columns. NULL means no value was recorded in that row; it is not a zero reading.

For current history, filter source to galaxy_watch and is_superseded to false. Keep replaced import revisions for history review.

The source column retains a legacy database default. Current ingestion explicitly supplies galaxy_watch, and measurement_origin distinguishes direct watch data from Health history imports.

A stress score is derived from HRV and personal references. It is experimental and not a medical-grade stress measurement.

Access: row-level security policies govern caregiver operations.

Table 28
DATA DICTIONARY OF WATCH LOCATION POINTS

public.watch_location_points. Keeps timestamped positions received from the watch for the movement trail and safe-zone evaluation. location_source distinguishes satellite GPS, fused and network positions.

Column Name	Data Type	U	Description	Nullable	PK	FK
accuracy_m	DOUBLE PRECISION	No	Reported horizontal position accuracy in meters.	No	No	No
device_id	UUID	*	Registered watch associated with this record.	No	No	Yes
elderly_id	UUID	No	Patient profile associated with this record.	No	No	Yes
id	UUID	Yes	Unique identifier for a received watch position.	No	Yes	No
latitude	DOUBLE PRECISION	No	Latitude in decimal degrees, from -90 to 90.	No	No	No
location_source	TEXT	No	Position provider: watch_gps, fused or network.	No	No	No
longitude	DOUBLE PRECISION	No	Longitude in decimal degrees, from -180 to 180.	No	No	No
measured_at	TIMESTAMPTZ	*	Original time the position was measured.	No	No	No
received_at	TIMESTAMPTZ	No	Time the backend received the watch telemetry.	No	No	No

Foreign keys: device_id references public.wearable_devices(id); elderly_id references public.elderly_profiles(elderly_id).

Unique combination: device_id, measured_at.

Safe-zone confirmation uses fresh, accurate watch_gps positions and a running monitoring session. Fused or network points may appear on the map but do not confirm a boundary crossing.

Access: caregiver reads are subject to patient ownership policies; writes use backend functions or triggers.

Table 29
DATA DICTIONARY OF MEDICATION SCHEDULES

public.medication_schedules. Stores patient medication plans, scheduled reminder times and the time zone used to calculate doses.

Column Name	Data Type	U	Description	Nullable	PK	FK
caregiver_id	UUID	No	Caregiver account associated with this record.	No	No	Yes
color_tag	TEXT	No	Display color selected for the medicine.	Yes	No	No
created_at	TIMESTAMPTZ	No	Date and time the database record was created.	Yes	No	No
dosage	TEXT	No	Dose amount or strength entered by the caregiver.	Yes	No	No
elderly_id	UUID	No	Patient profile associated with this record.	No	No	Yes
end_date	DATE	No	Date the schedule ends, if set.	Yes	No	No
form	TEXT	No	Medication form, such as tablet or liquid.	Yes	No	No
frequency	TEXT	No	How often the medicine should be taken.	No	No	No
id	UUID	Yes	Unique identifier for this medication schedule.	No	Yes	No
instructions	TEXT	No	Administration instructions.	Yes	No	No
is_active	BOOLEAN	No	Whether the record is currently active.	Yes	No	No
medication_name	TEXT	No	Name of the prescribed or tracked medicine.	No	No	No
prescribed_by	TEXT	No	Prescriber name, when provided.	Yes	No	No
reminder_enabled	BOOLEAN	No	Whether medication reminders are enabled.	Yes	No	No
start_date	DATE	No	Date the schedule begins.	No	No	No
times_of_day	TEXT[]	No	Array of daily reminder times, normally in HH:mm format.	Yes	No	No

Column Name	Data Type	U	Description	Nullable	PK	FK
timezone	TEXT	No	IANA time-zone name used to calculate reminder and dose times.	No	No	No
updated_at	TIMESTAMPTZ	No	Date and time the record was last updated.	Yes	No	No

Foreign keys: caregiver_id references public.caregivers(id); elderly_id references public.elderly_profiles(elderly_id).

Access: row-level security policies govern caregiver operations.

Table 30
DATA DICTIONARY OF MEDICATION LOGS

public.medication_logs. Records the outcome of an individual scheduled dose, including doses marked taken on the phone or watch.

Column Name	Data Type	U	Description	Nullable	PK	FK
created_at	TIMESTAMPTZ	No	Date and time the database record was created.	Yes	No	No
dose_key	TEXT	*	Stable per-dose key used to prevent duplicate phone or watch confirmations.	Yes	No	No
elderly_id	UUID	No	Patient profile associated with this record.	No	No	Yes
id	UUID	Yes	Unique identifier for this medication log.	No	Yes	No
notes	TEXT	No	Optional free-text notes.	Yes	No	No
recorded_by	UUID	No	Caregiver who entered the dose status.	Yes	No	Yes
schedule_id	UUID	*	Medication schedule being recorded.	No	No	Yes
scheduled_time	TIMESTAMPTZ	No	Planned dose date and time.	No	No	No
status	TEXT	No	Dose state, such as taken or missed.	Yes	No	No
taken_at	TIMESTAMPTZ	No	Time the dose was marked taken.	Yes	No	No
watch_device_id	UUID	No	Watch that marked the dose taken; blank for a phone-only confirmation.	Yes	No	Yes

Foreign keys: elderly_id references public.elderly_profiles(elderly_id); recorded_by references public.caregivers(id); schedule_id references public.medication_schedules(id); watch_device_id references public.wearable_devices(id).

Unique combination: schedule_id, dose_key only when (dose_key IS NOT NULL).

The unique schedule_id and dose_key combination applies only when dose_key is not NULL, preventing repeated confirmations of the same dose.

Access: row-level security policies govern caregiver operations.

Table 31
DATA DICTIONARY OF APPOINTMENTS

public.appointments. Stores scheduled care appointments and the information used by appointment reminders.

Column Name	Data Type	U	Description	Nullable	PK	FK
appointment_at	TIMESTAMPTZ	No	Scheduled appointment date and time.	No	No	No
caregiver_id	UUID	No	Caregiver account associated with this record.	No	No	Yes
created_at	TIMESTAMPTZ	No	Date and time the database record was created.	Yes	No	No
doctor_name	TEXT	No	Doctor named for this appointment.	Yes	No	No
duration_min	INTEGER	No	Planned length in minutes.	Yes	No	No
elderly_id	UUID	No	Patient profile associated with this record.	No	No	Yes
id	UUID	Yes	Unique identifier for this appointment.	No	Yes	No
location	TEXT	No	Appointment venue or address.	Yes	No	No
notes	TEXT	No	Optional free-text notes.	Yes	No	No
reminder_sent	BOOLEAN	No	Whether a reminder was sent.	Yes	No	No
status	TEXT	No	Appointment state.	Yes	No	No
title	TEXT	No	Short title shown in the app.	No	No	No
updated_at	TIMESTAMPTZ	No	Date and time the record was last updated.	Yes	No	No

Foreign keys: caregiver_id references public.caregivers(id); elderly_id references public.elderly_profiles(elderly_id).

Access: row-level security policies govern caregiver operations.

Table 32
DATA DICTIONARY OF CAREGIVER NOTES

public.caregiver_notes. Stores patient care notes and pinned information entered by the caregiver.

Column Name	Data Type	U	Description	Nullable	PK	FK
caregiver_id	UUID	No	Caregiver account associated with this record.	No	No	Yes
content	TEXT	No	Caregiver's note text.	No	No	No
created_at	TIMESTAMPTZ	No	Date and time the database record was created.	Yes	No	No
elderly_id	UUID	No	Patient profile associated with this record.	No	No	Yes
id	UUID	Yes	Unique identifier for this caregiver note.	No	Yes	No
is_pinned	BOOLEAN	No	Whether this note is pinned above others.	Yes	No	No
title	TEXT	No	Short title shown in the app.	Yes	No	No
updated_at	TIMESTAMPTZ	No	Date and time the record was last updated.	Yes	No	No

Foreign keys: caregiver_id references public.caregivers(id); elderly_id references public.elderly_profiles(elderly_id).
Access: row-level security policies govern caregiver operations.

Table 33
DATA DICTIONARY OF HEALTH ALERTS

public.health_alerts. Stores vital warnings, SOS alerts and safe-zone exit alerts. Reading, acknowledgment and resolution are separate states.

Column Name	Data Type	U	Description	Nullable	PK	FK
acknowledged_at	TIMESTAMPTZ	No	Server-recorded time the caregiver acknowledged the alert.	Yes	No	No
acknowledged_by	UUID	No	Authenticated user who acknowledged the alert.	Yes	No	Yes
ai_generated	BOOLEAN	No	Whether the stored alert was flagged as AI generated. Threshold and safe-zone rules are deterministic.	Yes	No	No
alert_type	TEXT	No	Kind of vital or care alert.	No	No	No
elderly_id	UUID	No	Patient profile associated with this record.	No	No	Yes
emergency_event_id	UUID	No	Linked SOS event, retained independently of inbox notifications.	Yes	No	Yes
id	UUID	Yes	Unique identifier for this health alert.	No	Yes	No
is_read	BOOLEAN	No	Whether the alert was marked read; this does not acknowledge it.	Yes	No	No
is_resolved	BOOLEAN	No	Whether the alert has been resolved.	Yes	No	No
message	TEXT	No	Explanation presented with the alert.	No	No	No
metric_unit	TEXT	No	Unit of the observed value.	Yes	No	No
metric_value	NUMERIC	No	Observed value that triggered the alert.	Yes	No	No
resolution_note	TEXT	No	Final caregiver action note, limited to 2,000 characters.	Yes	No	No
resolved_at	TIMESTAMPTZ	No	Server-recorded resolution time; final for the response workflow.	Yes	No	No
resolved_by	UUID	No	Caregiver who resolved the alert.	Yes	No	Yes
severity	TEXT	No	Alert urgency level.	No	No	No
threshold_value	NUMERIC	No	Comparison threshold used for the alert.	Yes	No	No
title	TEXT	No	Short title shown in the app.	No	No	No
triggered_at	TIMESTAMPTZ	No	Time the alert was triggered.	Yes	No	No

Foreign keys: acknowledged_by references auth.users(id); elderly_id references public.elderly_profiles(elderly_id); emergency_event_id references public.emergency_events(id); resolved_by references public.caregivers(id).
The response workflow progresses from New to Acknowledged to Resolved. Marking a notice read does not acknowledge or resolve its alert.
respond_to_care_alert records the responder and server time, requires a 3 to 2,000 character resolution note, and keeps the final response unchanged on retries.
Access: row-level security policies govern caregiver operations.

Table 34
DATA DICTIONARY OF ALERT RESPONSE EVENTS

public.alert_response_events. Keeps server-recorded acknowledgment and resolution events for an alert, including the responder, response time and final note.

Column Name	Data Type	U	Description	Nullable	PK	FK
action	TEXT	*	Response transition: acknowledged or resolved.	No	No	No
actor_id	UUID	No	Authenticated user responsible for the response; may become blank if the account is deleted.	Yes	No	Yes
alert_id	UUID	*	Health alert to which the response belongs.	No	No	Yes
elderly_id	UUID	No	Patient profile associated with this record.	No	No	Yes
id	UUID	Yes	Unique identifier for the recorded alert response event.	No	Yes	No
note	TEXT	No	Action note saved with resolution, up to 2,000 characters.	Yes	No	No
occurred_at	TIMESTAMPTZ	No	Server timestamp of the acknowledgment or resolution.	No	No	No

Foreign keys: actor_id references auth.users(id); alert_id references public.health_alerts(id); elderly_id references public.elderly_profiles(elderly_id).

Unique combination: alert_id, action.

The unique alert_id and action pair allows one acknowledgment and one resolution event per alert. Caregivers read the audit history; backend triggers create the events.

Access: caregiver reads are subject to patient ownership policies; writes use backend functions or triggers.

Table 35
DATA DICTIONARY OF CAREGIVER NOTIFICATIONS

public.caregiver_notifications. Stores the caregiver notification inbox and related push-delivery status. A notification may reference a saved health alert.

Column Name	Data Type	U	Description	Nullable	PK	FK
alert_id	UUID	No	Related health alert, if any.	Yes	No	Yes
body	TEXT	No	Full message shown to the caregiver.	No	No	No
caregiver_id	UUID	*	Caregiver account associated with this record.	No	No	Yes
channel	TEXT	No	Delivery channel, such as in-app.	Yes	No	No
data	JSONB	No	JSON payload containing routing, source-record and related care-event identifiers.	Yes	No	No
delivery_status	TEXT	No	Stored notification delivery state; not proof that a caregiver saw the message.	Yes	No	No
elderly_id	UUID	No	Patient profile associated with this record.	Yes	No	Yes
event_key	TEXT	*	Stable event identifier used to prevent duplicate notifications for a caregiver.	Yes	No	No
id	UUID	Yes	Unique identifier for this caregiver notification.	No	Yes	No
is_read	BOOLEAN	No	Whether the notification has been read.	Yes	No	No
notification_type	TEXT	No	Notification category.	No	No	No
push_error	TEXT	No	Latest recorded push-provider or delivery-processing error, if any.	Yes	No	No
push_status	TEXT	No	Push-registration or delivery-processing status; separate from inbox read state.	No	No	No
read_at	TIMESTAMPTZ	No	Time the notification was read.	Yes	No	No
sent_at	TIMESTAMPTZ	No	Time the notification was sent.	Yes	No	No
title	TEXT	No	Short title shown in the app.	No	No	No

Foreign keys: alert_id references public.health_alerts(id); caregiver_id references public.caregivers(id); elderly_id references public.elderly_profiles(elderly_id).

Unique combination: caregiver_id, event_key only when (event_key IS NOT NULL).

A unique caregiver_id and event_key combination prevents repeated notices when event_key is present. Push-provider acceptance is separate from human acknowledgment.

Push retries and device delivery tokens are maintained in supporting delivery tables outside this core dictionary.

Access: row-level security policies govern caregiver operations.

Table 36
DATA DICTIONARY OF EMERGENCY EVENTS

public.emergency_events. Records SOS requests and emergency response details, including any captured vital snapshot.

Column Name	Data Type	U	Description	Nullable	PK	FK
description	TEXT	No	Optional details of the event.	Yes	No	No
elderly_id	UUID	No	Patient profile associated with this record.	No	No	Yes
event_type	TEXT	No	Type of emergency or SOS event.	No	No	No
id	UUID	Yes	Unique identifier for this emergency event.	No	Yes	No
is_resolved	BOOLEAN	No	Whether the event has been resolved.	Yes	No	No
location	TEXT	No	Recorded location text, if available.	Yes	No	No
resolution_notes	TEXT	No	Notes about the event resolution.	Yes	No	No
responded_at	TIMESTAMPTZ	No	Time a caregiver responded.	Yes	No	No
responded_by	UUID	No	Caregiver who responded.	Yes	No	Yes
response_time_s	INTEGER	No	Elapsed response time in seconds.	Yes	No	No
severity	TEXT	No	Emergency urgency level.	No	No	No
triggered_at	TIMESTAMPTZ	No	Time the emergency was triggered.	Yes	No	No
vital_snapshot	JSONB	No	Vital values captured with the event.	Yes	No	No

Foreign keys: elderly_id references public.elderly_profiles(elderly_id); responded_by references public.caregivers(id).
health_alerts.emergency_event_id links an SOS event to its alert response. Clearing an inbox notification does not remove that relationship.
Access: row-level security policies govern caregiver operations.

Table 37
DATA DICTIONARY OF AI CHATBOT SESSIONS

public.ai_chatbot_sessions. Groups Elle conversation messages by caregiver and patient.

Column Name	Data Type	U	Description	Nullable	PK	FK
caregiver_id	UUID	No	Caregiver account associated with this record.	No	No	Yes
context_type	TEXT	No	Scope of care context for the conversation.	Yes	No	No
elderly_id	UUID	No	Patient profile associated with this record.	No	No	Yes
ended_at	TIMESTAMPTZ	No	Time the conversation ended.	Yes	No	No
id	UUID	Yes	Unique identifier for this Elle chat session.	No	Yes	No
is_active	BOOLEAN	No	Whether the conversation is still active.	Yes	No	No
session_title	TEXT	No	Title shown in chat history.	Yes	No	No
started_at	TIMESTAMPTZ	No	Time the conversation began.	Yes	No	No
total_messages	INTEGER	No	Number of stored messages in the session.	Yes	No	No

Foreign keys: caregiver_id references public.caregivers(id); elderly_id references public.elderly_profiles(elderly_id).
Access: row-level security policies govern caregiver operations.

Table 38
DATA DICTIONARY OF AI CHATBOT MESSAGES

public.ai_chatbot_messages. Stores chat messages and the source-reading references or aggregate-history evidence returned with an assistant response.

Column Name	Data Type	U	Description	Nullable	PK	FK
content	TEXT	No	Message text.	No	No	No
created_at	TIMESTAMPTZ	No	Date and time the database record was created.	Yes	No	No
evidence	JSONB	No	JSON array of source-reading references returned with the assistant reply; defaults to an empty array.	No	No	No

Column Name	Data Type	U	Description	Nullable	PK	FK
evidence_summary	JSONB	No	JSON object describing a factual history check, its scope and sample counts.	Yes	No	No
has_health_data	BOOLEAN	No	Whether the response used care or vital data.	Yes	No	No
id	UUID	Yes	Unique identifier for this Elle chat message.	No	Yes	No
is_flagged	BOOLEAN	No	Whether the message was flagged for review.	Yes	No	No
role	TEXT	No	Speaker role, such as user or assistant.	No	No	No
session_id	UUID	No	Conversation to which this message belongs.	No	No	Yes
tokens_used	INTEGER	No	Model token count, when recorded.	Yes	No	No

Foreign keys: session_id references public.ai_chatbot_sessions(id).

evidence items include reference, record_id, metric, label, value, unit, measured_at, source and origin. record_id is a JSON reference, not a declared foreign-key column.

evidence_summary records scope, timezone, from, until, generated_at, metric, sample_count, matching_count, threshold and comparison. The examples do not replace the complete aggregate count.

Access: row-level security policies govern caregiver operations.

Table 39
DATA DICTIONARY OF PATIENT SAFE ZONES

public.patient_safe_zones. Stores one configured safe-zone center and radius per patient, including whether boundary alerts are enabled.

Column Name	Data Type	U	Description	Nullable	PK	FK
caregiver_id	UUID	No	Caregiver account associated with this record.	No	No	Yes
elderly_id	UUID	Yes	Patient identifier and primary key; only one configured zone per patient.	No	Yes	Yes
enabled	BOOLEAN	No	Whether boundary checks are enabled; false means alerts are paused.	No	No	No
latitude	DOUBLE PRECISION	No	Latitude in decimal degrees, from -90 to 90.	No	No	No
longitude	DOUBLE PRECISION	No	Longitude in decimal degrees, from -180 to 180.	No	No	No
name	TEXT	No	User-defined place name, from 1 to 80 trimmed characters.	No	No	No
radius_m	INTEGER	No	Safe-zone radius in whole meters, from 100 to 5,000.	No	No	No
revision	UUID	No	Configuration version changed on saving, used to invalidate obsolete notifications.	No	No	No
updated_at	TIMESTAMPTZ	No	Date and time the record was last updated.	No	No	No

Foreign keys: caregiver_id references auth.users(id); elderly_id references public.elderly_profiles(elderly_id).

Changes are saved through save_patient_safe_zone after patient ownership is checked. Saving resets confirmation and changes the configuration revision.

An exit or return needs at least three qualifying watch GPS fixes spanning 20 seconds. Accuracy, boundary margins and monitoring status are checked by the backend; a missing fix does not prove the patient is inside.

Access: caregiver reads are subject to patient ownership policies; writes use backend functions or triggers.

Table 40
DATA DICTIONARY OF SAFE ZONE EVENTS

public.safe_zone_events. Keeps confirmed exits and returns with the zone configuration, measured distance and accuracy recorded at that event.

Column Name	Data Type	U	Description	Nullable	PK	FK
accuracy_m	DOUBLE PRECISION	No	Reported horizontal position accuracy in meters.	No	No	No
alert_id	UUID	No	Exit warning associated with this event; normally blank for a return.	Yes	No	Yes
created_at	TIMESTAMPTZ	No	Date and time the database record was created.	No	No	No
direction	TEXT	No	Confirmed boundary transition: outside or inside.	No	No	No

Column Name	Data Type	U	Description	Nullable	PK	FK
distance_m	DOUBLE PRECISION	No	Measured distance in meters from the configured zone center.	No	No	No
elderly_id	UUID	No	Patient profile associated with this record.	No	No	Yes
id	UUID	Yes	Unique identifier for a confirmed boundary event.	No	Yes	No
measured_at	TIMESTAMPTZ	No	Watch timestamp of the fix that completed boundary confirmation.	No	No	No
zone_name	TEXT	No	Snapshot of the place name when the boundary event occurred.	No	No	No
zone_revision	UUID	No	Snapshot of the zone configuration version used for this event.	No	No	No

Foreign keys: alert_id references public.health_alerts(id); elderly_id references public.elderly_profiles(elderly_id).

zone_revision and zone_name preserve the historical configuration. They are snapshots, not foreign keys to a previous configuration row.

A confirmed exit creates a health alert and notification. A confirmed return creates a boundary event and notification; it does not automatically resolve the earlier exit warning.

Access: caregiver reads are subject to patient ownership policies; writes use backend functions or triggers.